Assignment I S2 Mathematics and Physics 24MAT121 ODE

(Submit on or before 23/03/206)

1. A radioactive substance decays at a rate proportional to the amount present. If the initial amount is 100 g and after 5 hours it becomes 80 g, find the amount at time t.

2. Find a general solution.

   a. $y' = \sec^2 y$

   b. $xy' = x + y \quad (\text{Set } y/x = u)$

3. Solve the IVP.

   a. $xy' + y = 0, \quad y(4) = 6$

   b. $y' = 1 + 4y^2, \quad y(1) = 0$

4. Test for exactness. If exact, solve. If not, use an integrating factor as given or obtained by inspection or by the theorems in the text. Also, if an initial condition is given, find the corresponding particular solution.

   a. $e^{2x}(2\cos y\,dx - \sin y\,dy) = 0, \quad y(0) = 0$

   b. $(x^2 + y^2)\,dx - 2xy\,dy = 0$

5. Find the general solution. If an initial condition is give find also the corresponding particular solution and graph

   $xy' + 4y = 8x^4, \quad y(1) = 2$ Hint: Bernoulli equation

6. Reduce to first order and solve

   $x^2y'' - 5xy' + 9y = 0, \quad y_1 = x^3$

7. Solve the IVP.

   a. $y'' + 25y = 0, \quad y(0) = 4.6, \quad y'(0) = -1.2$

   b. $y'' + y' - 6y = 0, \quad y(0) = 10, \quad y'(0) = 0$

   c. $9y'' - 30y' + 25y = 0 \quad y(0) = 1, \quad y'(0) = 1$

8. A mass of 2 kg is attached to a spring with spring constant 18 N/m. The damping constant is 8 N·s/m. Initially the mass is pulled 0.2 m downward and released from rest. Find the equation of motion.

9. Solve the IVP.

$$x^2y'' + 3xy' + 0.75y = 0, \quad y(1) = 1,$$
$$y'(1) = -1.5$$

10 Solve the initial value problem.method of undetermined coefficients

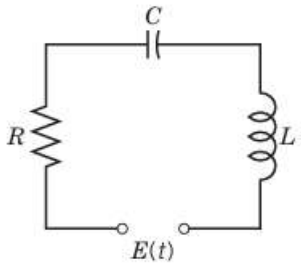
a. $y'' + 16y = 17e^x, \quad y(0) = 6, \quad y'(0) = -2$

b. $y'' - 3y' + 2y = 10 \sin x, \quad y(0) = 1, \quad y'(0) = -6$

11. Solve the given nonhomogeneous linear ODE by variation of parameters

$$x^2y'' - 2xy' + 2y = x^3 \sin x$$

12. Find the steady-state current in the *RLC*-circuit in Fig. 71 when $R = 50\ \Omega$, $L = 30$ H, $C = 0.025$ F, $E = 200 \sin 4t$ V.



13. Find a real general solution of the following systems.

$$y_1' = y_1 + y_2$$
$$y_2' = 3y_1 - y_2$$